AE 403W - GWAR Project 1: Proposal (Draft)

Luke Horton, Ethan Barichievich, Willie Peterson, Maxwell O'Neill

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In this report we will introduce our team to you and explain our Ground-Effect Vehicle (GEV) project for AE 403W. This report is written for writing consultants so we will provide clear definitions and figures that make the aerospace ideas easier to follow. We will first provide a brief context paragraph about who we are and what we are building. Then we will describe our current design idea and how we plan to test it.

Our team consists of Luke, Ethan, Willie, and Max, and we are all aerospace engineering students working on a GEV prototype design. This project will follow a typical engineering workflow: researching existing ground-effect vehicles, defining design requirements, selecting a configuration, manufacturing a prototype, testing it, and improving the design using the test results. In this report we will define acronyms the first time they appear For example as GEV (Ground-Effect Vehicle), CFD (Computational Fluid Dynamics), and RC (Radio Control).

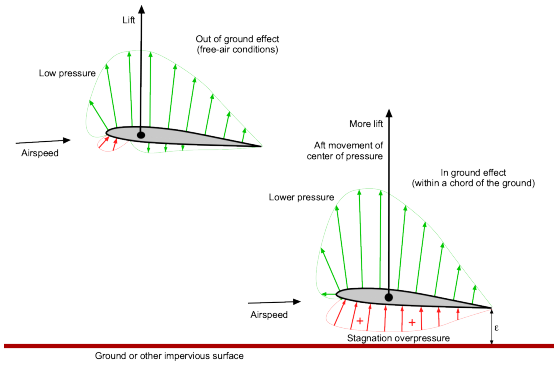

Figure 1: Pressure distribution and lift changes out of ground effect versus in ground effect near a surface.

Ground effect occurs when a wing travels very close to a surface and the airflow pattern changes compared to normal flight. As shown in Figure 1, the wing experiences increased pressure beneath it and reduced "losses" around the wing. This can increase lift and reduce induced drag. For our project the goal

is to demonstrate these efficiency gains at RC scale while maintaining stable and controllable flight near the water or ground. The main challenge is staying within an operating height where ground effect is strong especially with waves and spray that can disturb the vehicle.

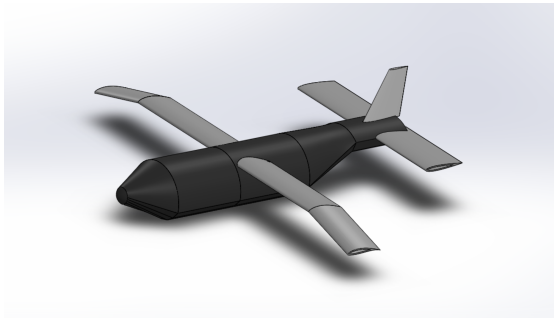

Figure 2: Current CAD concept of the ground-effect vehicle (GEV) prototype showing the fuselage, wing planform, and tail configuration.

Our current concept is a small-scale "flying yacht" GEV built using lightweight structural materials and manufacturing methods like 3-D printing. We selected the FX 63-137 airfoil because it delivers high lift at low speeds and has a proven track record in GEV designs. We are also considering distributed propulsion (four propellers) to improve low-speed thrust as well as control authority. This helps reduce redundancy for flight over the water. Figure 2 summarizes our initial wing sizing and geometry choices, which we selected to strengthen performance near the surface and reduce losses due to vortexes.

To evaluate performance we plan to combine simulations with real-world trials. Our goal during testing is to compare behavior in ground effect versus out of ground effect using measurements such as speed, ride height above the surface, and power draw. We will use controlled runs to see how small changes in height affect stability and efficiency, then use the data to guide design improvements. Based on our research we expect to iterate on wing geometry, weight placement, and flight controls to improve the crafts stability while also keeping the vehicle operating in the most efficient ground-effect region.